Design, Fabrication and Caractherization of All-Glass Laser-Written Vapor Cells for Integrated Photonics Applications

Rubén Piles Delgado

April 15, 2026

Contents

1 Introduction & Motivation

Sensors + MEMs + LWVCs + Alkali Gas + Why this technology is better

2 Design of Vapor Cells

2.1 Initial Considerations & Constraints

2.2 General Design of Individual LWVCs

2.3 Wafer-Level Approach

2.4 Real Examples

3 Fabrication Methods

3.1 Initial Considerations & Constraints

3.2 Types of Glasses & Quality Requirements

3.3 Femtolaser Laser

3.4 Optical Contact Bonding

3.4.1 Introduction

3.4.2 Chemical Cleaning & Activation

3.4.3 Plasma-Activated Bonding

Exposing glass surfaces to plasma —using a plasma cleaner— has proven to be a crucial step during the pre-bonding stage. Plasma (consisting of electrons, ions, free radicals, and photons) acts both as a cleaning and an activation agent in the context of glass-to-glass bonding and many other applications [1, 2, 3]. Figure 1 illustrates some of the many different processes that take place on the surface during plasma treatment; some of the most relevant ones are:

1. **Removal of contaminants from the surface:** Plasma removes contaminants from the surface —specially organic matter— and produces more dangling bonds, which leads to higher bonded areas.

2. **Increase in the number of silanol groups:** Plasma initiates a modification of the surface, increasing its number of silanol groups (Si-O-), which are directly correlated to bonding strength.

3. **Improvement of the diffusivity of water and gas trapped at the interface:** Plasma treatment increases diffusivity and thus enhances the evacuation of water and gas trapped between the two glass substrates during the bonding stage. Water evacuation is beneficial as it is needed to replace weak, reversible

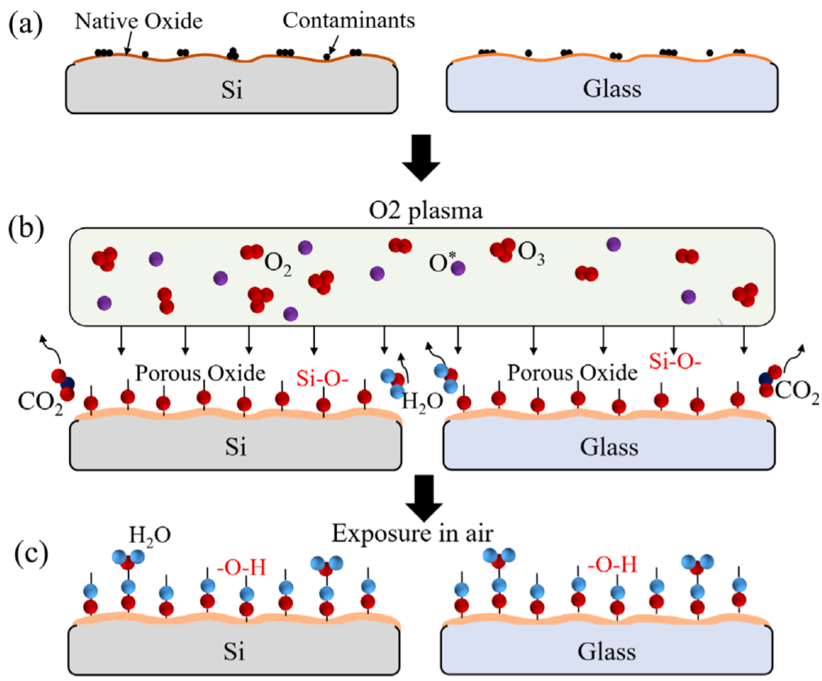

Figure 1: Effects of plasma treatment on silica surfaces. Plasma helps to remove contaminants while increasing the number of available silanol groups. Adapted from [4]

hydrogen bonds —which play a relevant role to start the bonding process— by stronger Si-O-Si bonds. Removal of trapped gas is also desirable, since it prevents the formation of bubbles during the annealing step.

Common gasses used in these plasma treatments are O_2, N_2, and Ar. When operating a plasma cleaner, special attention must be paid to its power, chamber pressure, gas flow, and operating times, as overexposure can lead to an increase of surface roughness and negatively affect the quality of the final bond.

Exposing plasma-treated surfaces to chemicals should be avoided, since these substances have the capacity to etch or alter the surface, effectively eliminating the benefits of this procedure [5]. Therefore, plasma treatment should be carried after all chemical cleaning or activation steps. Nevertheless, exposing plasma-treated surfaces to air or de-ionized water (DIW) has proven to be beneficial, since water molecules can attach to the Si-O- groups present on the surface and create silanol (Si-OH) groups, which are crucial to start the bonding process between two glass surfaces. Therefore, an effective workflow could be chemical cleaning and activation, followed by plasma treatment, a DIW rinse and finally proceeding with pre-bonding and annealing.

Plasma treatment becomes an essential tool in the context of low-temperature optical contact bonding, since it can yield strong bonding energies even at annealing temperatures as low as 100 °C [6].

3.5 Annealing

When two sufficiently flat and smooth glass surfaces are brought into contact, the direct bonding process initiates. One of the first mechanisms to occur is the formation of hydrogen bonds. These bonds can subsequently be transformed into stronger covalent bonds, as described by the following chemical reaction:

$$\mathrm{Si-OH+HO-Si \rightleftharpoons Si-O-Si+H_2O}. \tag{0.1}$$

To achieve permanent covalent bonding, water must be removed from the interface, rendering the aforementioned reaction irreversible [1]. This is accomplished through a thermal treatment known as *annealing*.

Annealing involves baking the bonded components in a controlled environment. The primary parameters influencing the outcome are the temperature ramp, the equilibrium temperature, and the annealing time.

The *temperature ramp* refers to the heating rate within the oven. This gradient should be kept as low as possible, as rapid temperature increases can induce thermal stress in the glass substrates. Such stress may compromise mechanical and optical performance, or even result in substrate fracture.

Regarding *equilibrium temperatures*, higher values facilitate greater water diffusivity, thereby removing moisture from the interface more rapidly. Furthermore, increased viscous flow enhances surface contact [1]. However, high temperatures are often unsuitable for precision optics, as they may damage optical coatings or integrated electronics [7]. In such cases, plasma activation is essential, as it increases surface energy and water diffusivity at significantly lower temperatures [8]. This approach is referred to as *Low-Temperature Direct Bonding*.

Concerning *annealing times*, literature indicates that bonding energy increases monotonically over time, whether through a dedicated annealing process or by leaving surfaces in contact at room temperature. The relationship between bonding energy and time follows a logarithmic trend:

$$\gamma \propto \ln\left(\frac{t}{\mu}\right), \tag{0.2}$$

where t is the annealing time and μ represents a time constant. The role of annealing is effectively to decrease this μ constant, allowing maximum bonding energy to be reached much faster. Typical annealing times for optical applications range from 2 to 24 hours [8, 4, 7, 9, 10]. An example of how the bonding energy evolves with time can be seen in Figure 2.

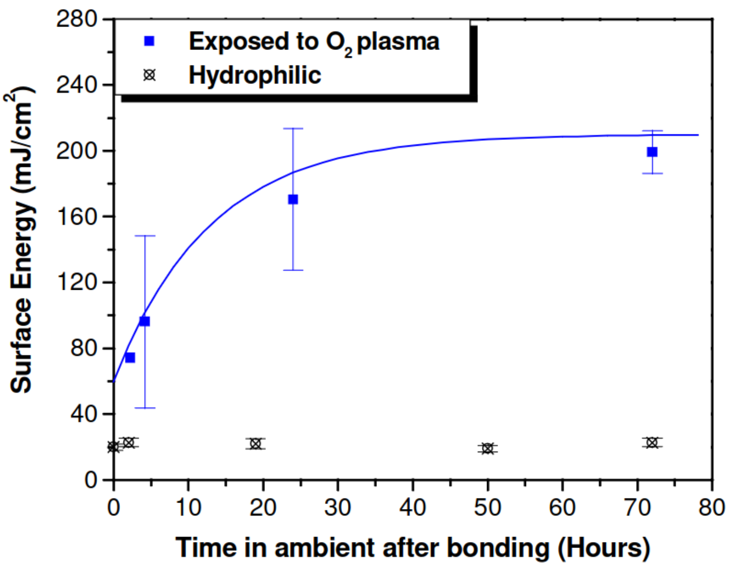

Figure 2: Bonding energy evolution with time. From [8]

3.5.1 Annealing pressure

In the context of optical contact bonding, mechanical pressure applied during the annealing phase acts as a vital catalyst by facilitating the deformation of surface asperities to increase the degree of intimate contact [11, 10], as displayed in Figure 4

This mechanical assistance accelerates the kinetics of silanol polymerization, allowing for the formation of covalent siloxane bonds at lower temperatures and elevating bond energy from typical room-temperature levels ($<0.5\,\mathrm{J\,m^{-2}}$) to values exceeding $2\,\mathrm{J\,m^{-2}}$, which often surpasses the bulk glass fracture toughness [12].

However, pressure must be carefully optimized; for instance, research on Borofloat 33 wafers indicates that a load of 0.07 MPa maximizes flexural strength, pressures beyond this limit can lead to stress concentrations and premature failure [10]. Furthermore, non-uniform pressure distributions introduce significant risks to device performance, such as stress-induced birefringence and permanent wavefront distortion [13, 14], necessitating well-planarized pressure plates —with peak-to-valley flatness in the order of micrometers— made of highly stiff materials —such as technical ceramics— [15] for high-precision optical and MEMS applications .

In the setup described by Birckigt et al. [15], pressure is used to enhance contact front propagation during the initial bonding phase. Convex and concave pressure plates —with waviness of 5 µm and 3 µm, respectively— have been employed, as shown in Figure 3. This setup mitigates the risk of creating interface voids and introduces radially symmetric residual stresses.

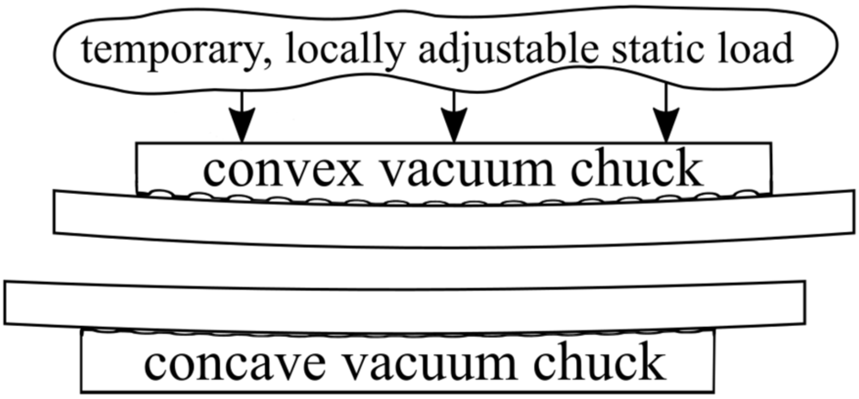

Figure 3: Setup described by Birckigt et al. [15]

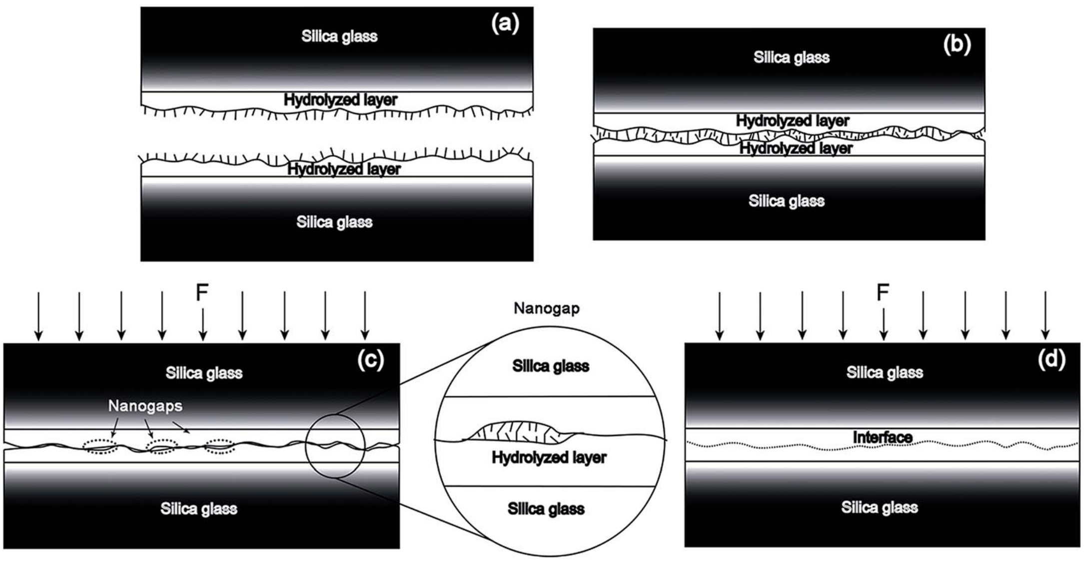

Figure 4: Effects of pressure in the bonding process. Interface voids can be prevented by applying forces in the direction perpendicular to the interface. From [11]

3.6 Other bonding methods

3.6.1 Anodic Bonding

3.7 Post-Treatments (Coatings, activation)

3.8 Step-by-Step Experimental Procedure

4 Carachterization and Quality Control

4.1 Strength Tests

4.2 Surface inspection using an Atomic Force Microscope

Atomic Force Microscopy (AFM) is a sophisticated type of scanning probe microscopy, which consists of a physical probe that scans the specimen in order to produce images of surfaces. Atomic Force Microscopes can map the topology of a surface with sub-nanometric resolution, which makes them an ideal tool for surface inspection in the context of atomic sensors, and more specifically in the context of Optical Contact Bonding, due to the strict glass roughness requirements of this method.

The AFM consists of a cantilever with a sharp tip at its end —with a radius of curvature on the order of nanometers, as shown in 5— that is used to scan the specimen surface. When the tip is brought near a sample, forces between them lead to a deflection of the cantilever according to Hooke's law. Forces that are measured in AFM include mechanical contact force, van der Waals forces and capillary forces, among others. Operating modes are divided into static *contact* modes and *tapping* modes, where the cantilever is vibrated at a given frequency.

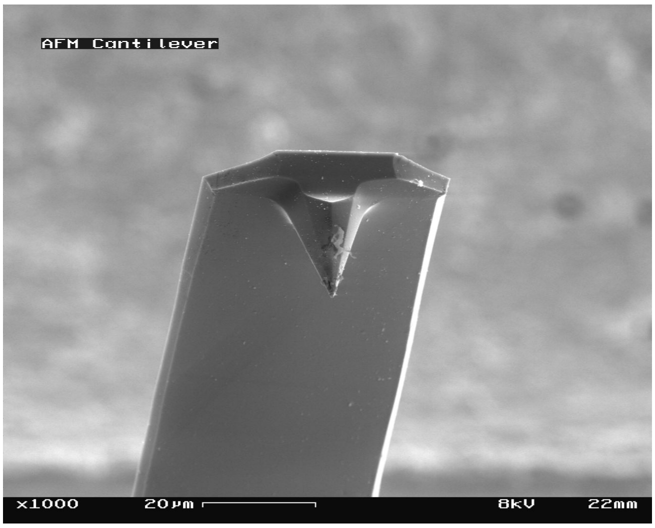

Figure 5: View of cantilever in Atomic Force Microscope (magnification 1000x) [16]

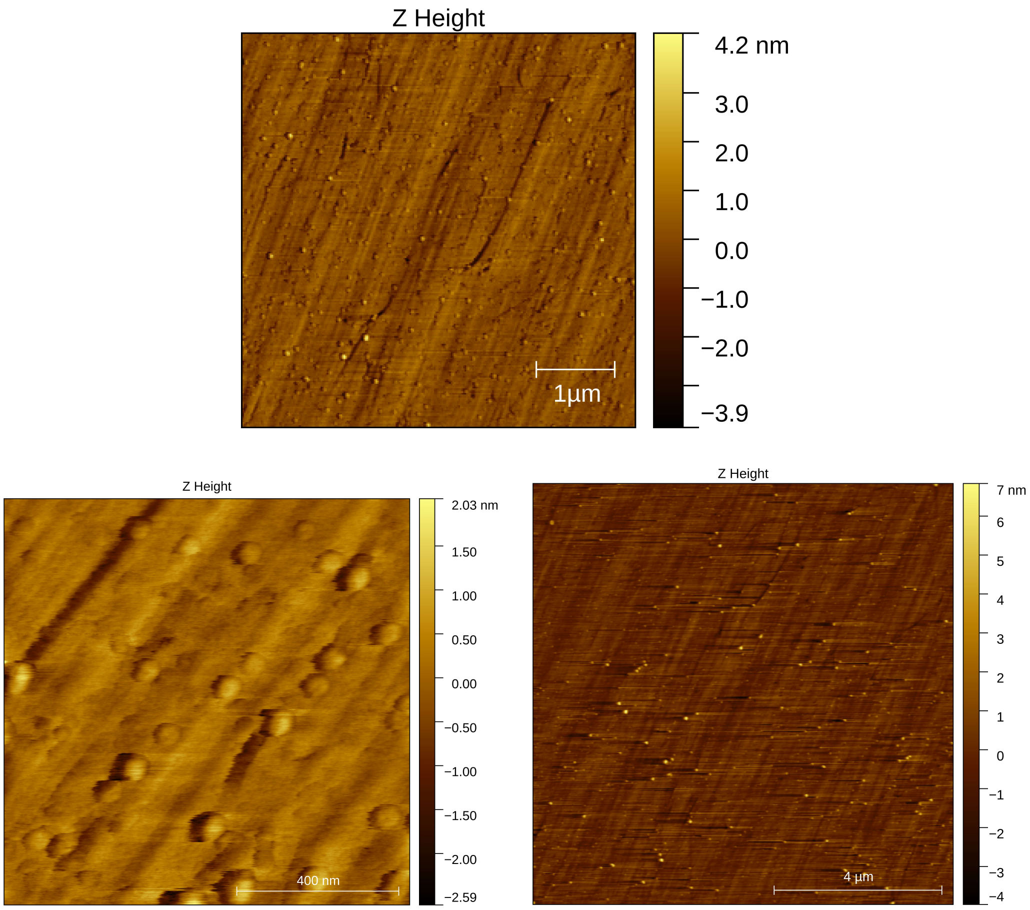

Figure 6: 2D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

A brand-new Nanoquartz glass disk, with specified Root Mean Square roughness of 500 pm and Total Thickness Variation (TTV) of 2 µm was inspected using a Park NX10 AFM, with a tip Park Scout 350 in *tapping* mode (resonance frequency 320 kHz, rate 0.5 Hz). It was examined in three different spots of its surface. The obtained data was analyzed using the *Gwyddion* software, where two 6 and three-dimensional 7 AFM images of the samples were obtained, along with the density distribution of roughness —displayed in 8. A detailed list of the estimated surface parameters can be found in 1. The precise definition of these parameters can be found in [17].

The results confirm that the Root Mean Square roughness of the disk is around 500 pm, as suggested by the manufacturer. An alternative estimation of the surface roughness, known as *Mean roughness* (Sa) was found to be in the same order of magnitude. Additionally, it was determined that the total height (S_q) is in the range of a few nanometers. The mean height is approximately zero, suggesting that

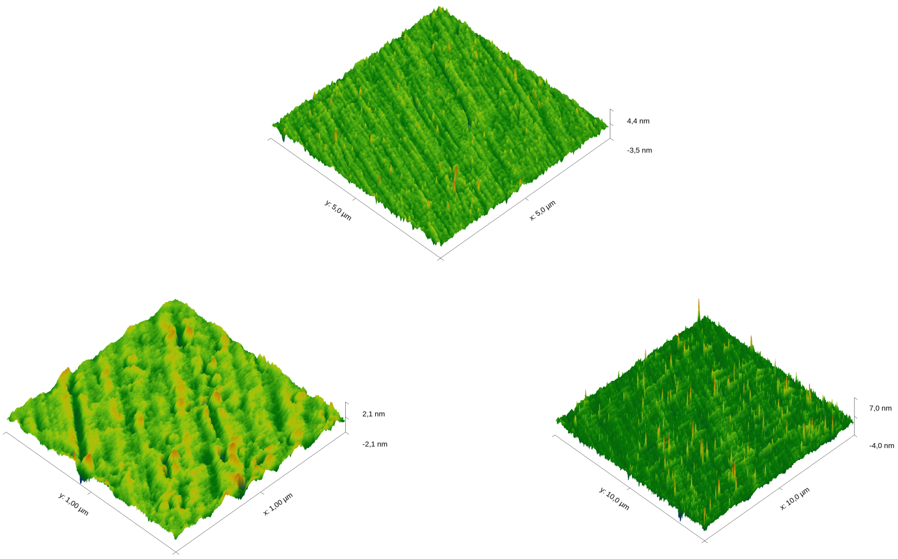

Figure 7: 3D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

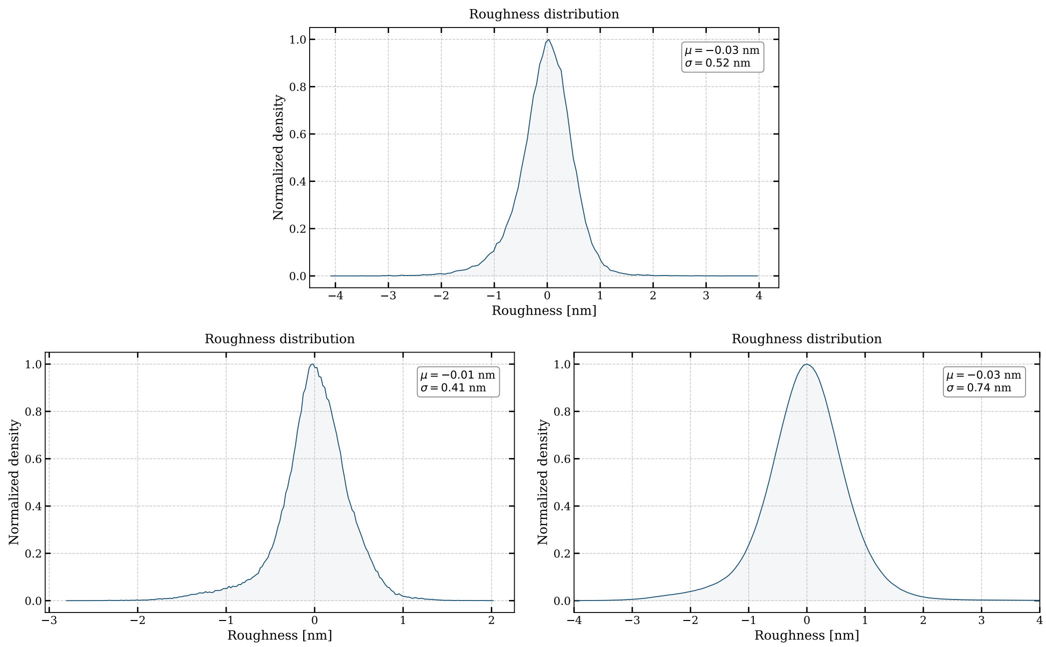

Figure 8: Surface roughness distribution obtained through AFM. It can be seen that most of the defects are below the 1 nm range. Furthermore, the mean (μ) is close to zero, confirming that the surface is uniform.

Sample	S_q [pm]	S_a [pm]	S_p [nm]	S_v [nm]	S_z [nm]	Mean height [nm]
1	518	387	4.02	-4.14	8.17	0.03
2	411	298	2.04	2.83	4.87	0.02
3	726	529	7.66	4.05	11.71	0.02

Table 1: Surface parameters obtained through AFM in three different parts of the surface: *Root mean square roughness* (S_q), *Mean roughness* (S_a), *Maximum peak height* (S_p), *Minimum valley depth* (S_v), *Maximum height* (S_z) and *Mean height*

the surface's defects are uniformly distributed. The outstanding surface parameters observed in this disk comply with Optical Contact Bonding requirements and therefore it is suitable for the fabrication of Laser-Written Vapor Cells and high-precission atomic sensors.

4.3 Leak Tests using Spectroscopy

Once the cell is sealed and activated with the alkali metal vapor (typically Rb or Cs), leak tests can be performed over time to estimate its hermeticity. Leak tests involving laser absorption spectroscopy have been reported in literature [18, 19, 20, 21, 22]. In these tests, a tunable laser interacts with the alkali atoms. By measuring the width and central frequency of the resulting absorption profile, the internal environment of the cell can be monitored.

The optical transitions of alkali metals are characterized by the so-called D-line doublet. This fine structure arises from the spin-orbit interaction, which splits the first excited state (nP) into two distinct energy levels: $n^2P_{1/2}$ and $n^2P_{3/2}$. The transition from the ground state $n^2S_{1/2}$ to the lower excited state $n^2P_{1/2}$ is known as the D1 line, while the transition to the higher state $n^2P_{3/2}$ is the D2 line. These transitions are illustrated in Figure 9.

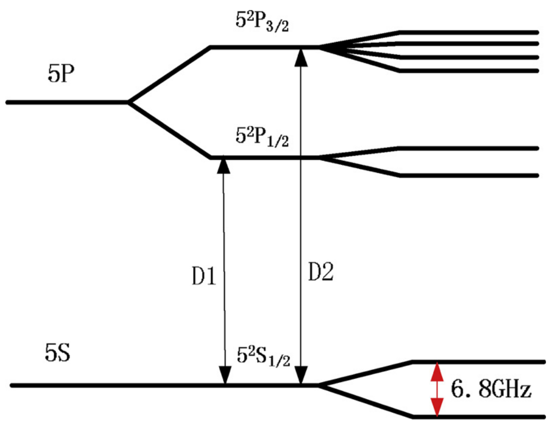

Figure 9: Energy level diagram showing the D1 and D2 transitions in an alkali metal atom. From [22]

Collisions between the alkali atoms and other gas molecules perturb the atomic energy levels, leading to collision-induced broadening and a frequency shift. If the cell's hermeticity is compromised and atmospheric gases leak into the cell, the resulting increase in internal pressure (P) will cause a detectable width broadening and a frequency shift of the Lorentzian profile. To quantify the gas pressure, the following relations are used:

$$\delta\nu = \delta P \tag{0.3}$$

$$\Delta\nu_L = \gamma P + \gamma_N \tag{0.4}$$

The first equation relates the frequency shift ($\delta\nu$) to the pressure (P) through the shift rate constant (δ). The second equation links the pressure to the Lorentzian full width at half maximum ($\Delta\nu_L$), involving the broadening rate constant γ and the natural linewidth γ_N.

Both δ and γ depend on the specific alkali-gas pair and the temperature, with values documented in the literature [18, 19, 20, 21, 22]. Furthermore, the broadening rate γ at a temperature T_2 can be estimated from a known value at T_1 using:

$$\gamma(T_2) = \gamma(T_1)\left(\frac{T_1}{T_2}\right)^{\frac{1}{2}} \tag{0.5}$$

Note that the shift constant δ does not follow such a simple scaling law due to its more complex temperature dependence [18]

4.3.1 Experimental Proposal

The spectroscopic framework described above can be directly applied to assess the hermeticity of the fabricated vapor cells. Here, an experimental procedure similar to the one followed by Maurice et al. [23] is described.

First, the sealed and activated cell is heated to a nominal operating temperature T_{op} —chosen so that the alkali vapor pressure is sufficient to produce a measurable absorption signal— and held there throughout the measurement campaign using a temperature-stabilized mount. Temperature stability is critical, since any drift in T_{op} would introduce a systematic error in the extracted pressure through the temperature dependence of γ and δ [18].

A tunable laser is scanned across the D1 or D2 absorption line and the transmitted intensity is recorded with a photodetector. To avoid power broadening, the probe beam intensity is kept well below the saturation intensity of the transition. The resulting spectrum is fitted to a Voigt profile, from which the Lorentzian width $\Delta\nu_L$

and the line center ν_0 are extracted. Since the Doppler width depends only on T_{op} and the atomic mass, the Lorentzian component is unambiguously separated. The internal pressure is then retrieved using Equations 0.3 and 0.4, evaluated at T_{op} using literature values of γ and δ [18, 19, 20, 21, 22], scaled if necessary via Equation 0.5. The two estimates of P —one from the width and one from the frequency shift— serve as a mutual consistency check.

Spectra are recorded at regular time intervals: initially daily, and subsequently weekly once short-term stability has been confirmed. The pressure $P(t)$ is plotted as a function of time and, in the case of a small but non-zero leak, a linear fit to $P(t)$ yields the leak rate $\dot{P}$. If no detectable increase is observed, the measurement noise propagated through Equation 0.4 sets an upper bound on the leak rate:

$$\sigma_P = \frac{\sigma_{\Delta\nu_L}}{\gamma(T_{\mathrm{op}})} \tag{0.6}$$

where $\sigma_{\Delta\nu_L}$ is the uncertainty on the fitted Lorentzian width. This bound provides a physically meaningful hermeticity statement even in the absence of a detectable leak, and allows direct comparison with benchmarks from the literature: the 707-day vacuum integrity demonstrated by Artusio-Glimpse et al. [24] using Rydberg EIT spectroscopy, and the short-term seal integrity inferred from CPT resonance stability by Kobtsev et al. [25] over timescales up to $\sim$1000 s.

4.4 Enhanced Helium Hermeticity in Micro-fabricated Vapor Cells

Helium plays a critical role in the operation of various atomic sensors, serving primarily as a working medium or buffer gas in high-sensitivity applications (e.g., magnetoencephalography [26]). However, due to its extremely small atomic radius, helium exhibits one of the highest permeation rates through glass walls compared to other gases and alkali metals typically confined within micro-electromechanical systems (MEMS) vapor cells.

For high-precision applications where long-term cell hermeticity is paramount, aluminum oxide (Al_2O_3, or alumina) has been identified as a highly effective barrier [27, 28]. Researchers have found that utilizing aluminosilicate glass—a glass matrix enriched with Al_2O_3—or depositing thin Al_2O_3 coatings directly onto standard glass cells significantly mitigates helium diffusion.

As illustrated in Figure 10, the permeability constant (K) of aluminosilicate glass is between two and three orders of magnitude lower than that of conventional borosilicate glass. Furthermore, applying an Al_2O_3 coating to borosilicate cells can independently reduce helium permeability by one to two orders of magnitude. Synergistically combining an aluminosilicate glass substrate with the aforementioned Al_2O_3 coating yields the best results, achieving ultra-low permeabilities on the order of 10^{-22} $\mathrm{m^2s^{-1}Pa^{-1}}$.

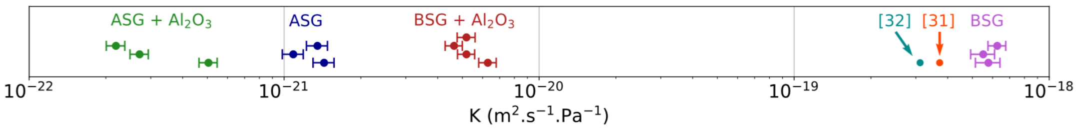

Figure 10: Comparison of the permeability constant K across uncoated borofloat, uncoated aluminosilicate, Al_2O_3-coated borofloat, and Al_2O_3-coated aluminosilicate glasses. From [27].

5 Experiments & Applications

5.1 Magnetometry Experiments on Vapor Cells

In this experiment, the goal was to demonstrate the capacity of vapor cells to serve as precise magnetometers in a laboratory setting. As described in [29, 30], when circularly polarized light interacts with alkali atoms (such as Rubidium or Cesium), it *transfers* some of its angular momentum into the gas. This effectively generates a non-zero electron spin polarization state inside the vapor cell.

In the presence of an external magnetic field $\vec{B}$, the atomic spins undergo Larmor precession at the frequency:

$$\omega_L = \gamma B \tag{0.7}$$

where γ is the gyromagnetic ratio (2π 3.5 kHz/G for Cesium-133). This precession modulates the projection of the atomic spin along the probe beam direction, causing periodic variations in the optical absorption and thus in the transmitted light intensity detected by the photodiode. When a linear magnetic field ramp —such as $B(t) = B_0 + \alpha t$— is applied across the vapor cell, the transmitted light intensity exhibits a Lorentzian resonance peak centered at $B = 0$. This phenomenon, known as *Hanle resonance*, arises from the following mechanism:

For $B \neq 0$, the Zeeman sublevels ($m_F = -F, -F+1, \ldots, +F$) are non-degenerate and precess at different rates, leading to dephasing of the atomic coherences. This dephasing destroys the spin polarization created by optical pumping, resulting in increased absorption and reduced transmission.

At $B = 0$, all Zeeman sublevels become degenerate, eliminating the precession-induced dephasing. The atomic coherences remain phase-locked, maintaining maximal spin polarization in the direction of the pump beam. Atoms in this polarized state exhibit reduced absorption for the circularly polarized probe light, producing a transmission maximum.

By analyzing the Lorentzian profile, properties of the vapor cell, such as the coherence lifetime (τ), can be determined. The relationship between the Lorentzian full width

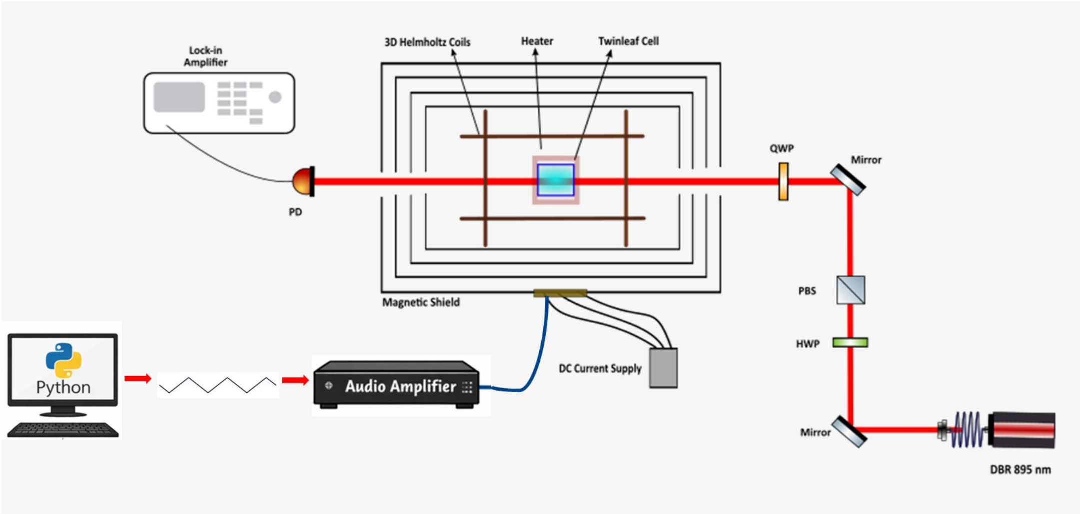

Figure 11: Experimental setup used to detect Hanle resonance

at half maximum (FWHM) and τ is given by:

$$\Delta B_{\mathrm{FWHM}} \sim \frac{1}{\gamma\tau} \tag{0.8}$$

5.1.1 Experimental Setup

A Twinleaf Vapor Cell, filled with Cesium and Nitrogen as buffer gas, was placed inside a magnetic shield to isolate it from background noise (e.g., Earth's magnetic field). A laser tuned to the D1 transition of Cs —λ =895 nm— is directed into the vapor cell. To control the power of the incoming light, a half-wave plate (HWP) and a polarizing beam splitter (PBS) were employed. Just before entering the magnetic shield, a quarter-wave plate (QWP) is placed to generate a circularly polarized beam.

Inside the magnetic shield, coils are used to generate an alternating magnetic field ramp ($B_x = B_0 + \alpha t$) along the beam axis (X-axis). As explained previously, this time-dependent field generates a time-dependent absorption profile, which is detected by a photodetector placed outside the magnetic cavity and connected to a lock-in amplifier. To correct for background noise, DC fields are applied along the Z- and Y-axes. This experiment was carried at room temperature. An illustration of this setup is displayed in Figure 11.

5.1.2 Results and Analysis

Measurements were made using different DC field configurations to study their effects on Lorentzian peak height, width, and noise. Figure 12 compares the fitted Lorentzian profiles detected by the photodetector for different DC field configurations.

From these results, it can be deduced that the chosen DC field configuration greatly affects the sharpness of the observed Hanle resonance. Signal amplitudes below 10 mV are obtained for the least optimal configuration ($B_y = 3600\,\mathrm{nT}$, $B_z = 0\,\mathrm{nT}$), while peaks greater than 40 mV are achieved with the configuration $B_y = 0\,\mathrm{nT}$, $B_z = 4815\,\mathrm{nT}$.

In conclusion, for this specific experimental setup, applying a DC field in the Y direction reduces the amplitude and increases the FWHM of the Hanle resonance, whereas a DC field in the Z direction has the opposite effect.

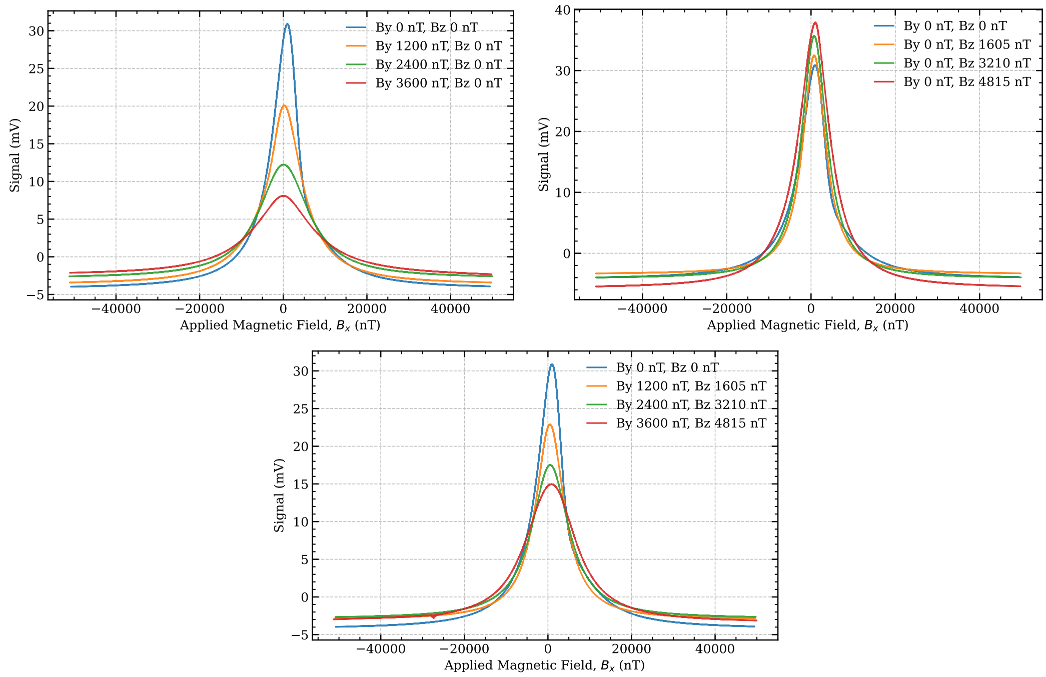

Figure 12: Lorentzian profiles when (a) B_z is varied and $B_y = 0$, (b) B_y is varied and $B_z = 0$, and (c) B_y and B_z are varied simultaneously

Since the goal of this experiment is to study Hanle resonances, the configuration $B_y = 0\,\mathrm{nT}$, $B_z = 4815\,\mathrm{nT}$ was further analyzed. Figure 13 shows the raw data and Lorentzian fit for this configuration. The amplitude of the resonance signal is 42.2 mV. It can be observed that the Lorentzian's maximum is not located exactly at the origin, but is displaced slightly to the right ($B_x = 865.5\,\mathrm{nT}$). This effect is observed systematically and can be attributed to background noise and electronic effects. A FWHM of 8726.3 nT is estimated. From this value, and using 0.8, the relaxation time of this cell at room temperature is ∼50 ms.

Experiments + magnetometry + space applications + clocks + integration + future applications

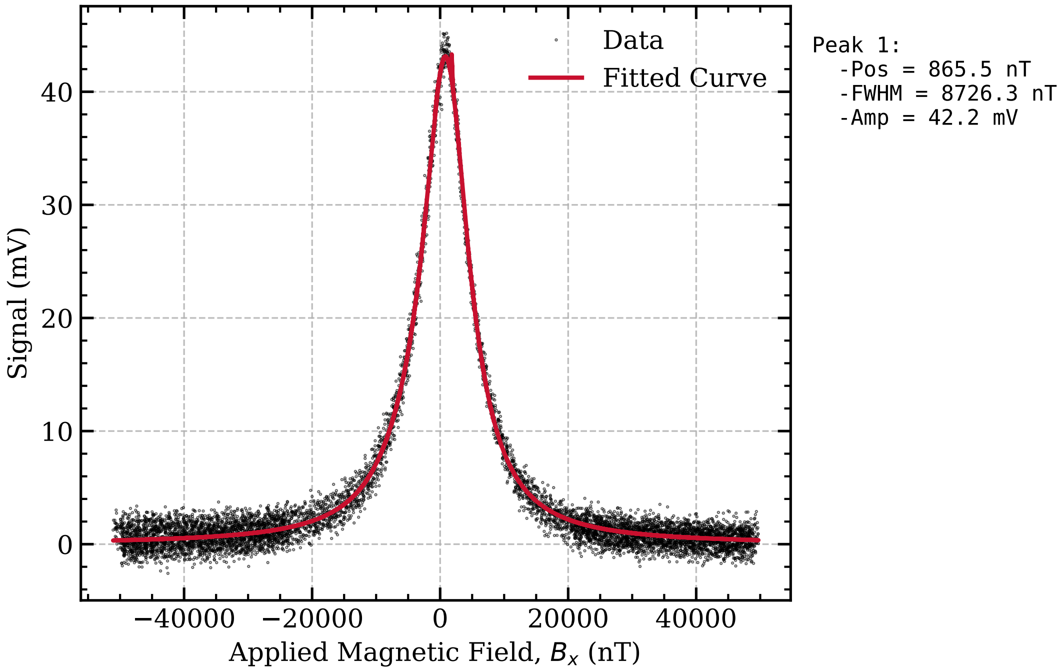

Figure 13: Raw data and Lorentzian fit of the Hanle resonance obtained for the optimal DC field configuration ($B_y = 0\,\mathrm{nT}$, $B_z = 4815\,\mathrm{nT}$)

6 Overlook & Conclusions

7 Appendix

7.1 Recipes in literature

Table 2: Summary of chemical cleaning procedures, surface activation methods, and resulting bonding qualities.

Reference	Chemical Cleaning Steps							Surface Activation	Post-Treatment	Bonding Strength
	1	**2**	**3**	**4**	**5**	**6**	**7**			
Domenico Tuli Thesis	DIW 10' 25°C	—	—	—	—	—	—	—	—	0.06 J/m^2
	Acetone 10' 25°C	—	—	—	—	—	—	Plasma 30s 15W 35mTorr	—	0.06 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Micro Soap 10' 25°C	DIW 10' 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	RCA1 10' 60°C	DIW 10' 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	RCA1 10' 60°C	NH_4OH 2' 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Micro Soap 10' 25°C	RCA1 10' 60°C	DIW 10' 25°C	—	—	—	—	0.06 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Micro Soap 10' 25°C	RCA1 10' 60°C	DIW 10' 25°C	—	—	—	—	0.1 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Micro Soap 10' 25°C	RCA1 10' 60°C	NH_4OH 2' 25°C	—	—	—	—	0.1 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Piranha 10' 60°C	DIW 10' 25°C	—	—	—	Plasma 30s 15W 35mTorr	—	0.1 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Piranha 10' 60°C	DIW 10' 25°C	Micro Soap 10' 25°C	RCA1 10' 60°C	DIW 10' 25°C	—	—	0.1 J/m^2
	Acetone 10' 25°C	DIW 10' 25°C	Piranha 10' 60°C	DIW 10' 25°C	Micro Soap 10' 25°C	RCA1 10' 60°C	NH_4OH 2' 25°C	—	—	0.1 J/m^2
Cho et al. 2001	HF (dipped)	Piranha 15'	RCA1 15'	—	—	—	—	O_2 Plasma 15-120s 50-300W	Annealing 50h >105°C	2.5 J/m^2
Eichler et al. 2008	RCA1	RCA2	—	—	—	—	—	O_2 Plasma 40s 30W 1atm	Annealing 6h 120°C	2.2 J/m^2
Li et al. 2013	SC1 10' 80°C	SC2 10' 80s	DIW (rinse)	RCA1 10' 60°C	—	—	—	O_2 Plasma 60s 50W 20mTorr	Press. 3x10^5 Pa → Annealing 6h 200-800°C	11 MPa BS
Wang et al. 2020	Acetone	Ethanol	DIW	—	—	—	—	O_2 Plasma 90s 200W 375mTorr	Wait 12h Room T → Annealing 12h 150°C	5.5 MPa BS
Takei et al. 2010	RCA1 10' 70°C	DIW (rinse)	HF 2%	—	—	—	—	O_2 Plasma 30s 400W 900mTorr	Annealing 8h 250°C 5MPa	1 MPa BS
Lee et al. 2022	—	—	—	—	—	—	—	N_2 Plasma 15s 100W 100mTorr	DIW → Annealing 1-4h 200-500°C	2.5x w/o plasma
Suni et al. 2002	—	—	—	—	—	—	—	N_2 Plasma 30s 50-150W	RCA1 → DIW 5' → Annealing → Dicing → Annealing	>2.5 J/m^2
	—	—	—	—	—	—	—	O_2 Plasma 45s 50-150W	Annealing 2h 100°C → Dicing → Annealing 2h 200°C	~2.5 J/m^2
Park et al. 2025	Piranha	HF dip	—	—	—	—	—	O_2 Plasma 30s 800W 800mTorr	Press. 5' 5000mBar → Annealing 1h 400°C	~1.7 J/m^2
Our proc. (P11-P13)	Acetone 10' 25°C	DIW 5' + dry	Piranha 10' 75°C	DIW 5' + dry	RCA1 10' 75°C	DIW 5' + dry	—	O_2 Plasma 60s 30W 1200mTorr O_2 Plasma 120s 30W 1200mTorr O_2 Plasma 180s 30W 1200mTorr	DIW 5' → Annealing >12h 150°C	TBD
Our proc. (P21-P23)	Acetone 10' 25°C	DIW 5' + dry	Piranha 10' 75°C	DIW 5' + dry	—	—	—	O_2 Plasma 60s 30W 1200mTorr O_2 Plasma 120s 30W 1200mTorr O_2 Plasma 180s 30W 1200mTorr	DIW 5' → Annealing >12h 150°C	TBD

Bibliography

[1] R. et al., “Handbook of wafer bonding,” 2010.

[2] M. et al., “A review of hydrophilic silicon wafer bonding,” 2014.

[3] T. Sumi, “Phd thesis: Direct wafer bonding for mems and microelectronics,” 2006.

[4] Y. et al., “Plasma-activated silicon–glass high-strength multistep bonding for low-temperature vacuum packaging,” 2023.

[5] N. et al., “Atmospheric pressure dielectric barrier discharge plasma-enhanced optical contact bonding of coated glass surfaces,” 2021.

[6] M. E. et al., “Atmospheric-pressure plasma pretreatment for direct bonding of silicon wafers,” 2008.

[7] L. et al., “Low temperature si/si wafer direct bonding using a plasma activated method,” 2013.

[8] C. et al., “Low temperature si direct bonding by plasma activation,” 2001.

[9] P. et al., “Wafer direct bonding: Tailoring adhesion between brittle materials,” 1998.

[10] T. et al., “Glass-to-glass fusion bonding quality and strength evaluation with time, applied force, and heat,” 2022.

[11] M. et al., “Low temperature direct bonding of silica glass via wet chemical surface activation,” 2015.

[12] K. et al., “Glass-glass direct bonding,” 2014.

[13] B. et al., “Effect of clamping force on distortion of the optical surface of monochromators during assembly,” 2022.

[14] W. et al., “Wavefront measurement techniques used in high power lasers,” 2014.

[15] P. Birckigt, K. Grabowski, G. Leibeling, T. Flügel-Paul, M. Heusinger, H. Ouslimani, and S. Risse, “Effects of static load and residual stress on fused silica direct bonding interface properties,” *Appl. Phys. A*, 2021.

[16] S. (Wikipedia), “Afm cantilever image,” 2008.

[17] I. O. for Standarization(ISO), “Iso 25178 - geomertical product specifications (gps),” 2012.

[18] P. et al., “Pressure broadening and shift of the cesium d1 transition.”

[19] ——, “Pressure broadening and shift of the cesium d2 transition.”

[20] A. et al., “High-resolution measurement of the pressure broadening and shift of the cs d1 and d2 lines by n2 and he buffer gasses.”

[21] P. et al., “Pressure broadening and shift of the rubidium d1 transitionand potassium d2 transitions by various gases.”

[22] S. et al., “Broadening and shifting of rb d1 and d2 line in low pressure vapor cell.”

[23] M. et al., “Wafer-level vapor cells filled with laser-actuated hermetic seals for integrated atomic devices,” 2022.

[24] A.-G. et al., “Wafer-level fabrication of all-dielectric vapor cells electrometry.”

[25] K. et al., “Cpt atomic clock with cold-technology-based vapour cell.”

[26] L. et al., “Magnetoencephalography with optically pumped he-4 magnetometers at ambient temperature,” 2019.

[27] C. et al., “Reduction of helium permeation in microfabricated cells using aluminosilicate glass substrates and al2o3 coatings,” 2023.

[28] D. et al., “Low helium permeation cells for atomicmicrosystems technology,” 2016.

[29] L. et al., “Laser-written vapor cells for chip-scale atomicsensing and spectroscopy,” 2022.

[30] T. et al., “Optically pumped magnetometers: From quantum origins to multichannel magnetoencephalography,” 2019.